

Decision support for farmers – a systems approach illustrated by an analysis on the impact of integrating chickens into German apple orchards

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Abstract

Agroforestry practices are widely considered to have numerous benefits compared to conventional farming on e.g. income stability, ecosystem services and food security, to name just a few. However, there are a range of hurdles linked to integrating tree and livestock systems into agriculture. These include aspects of labour intensity, investment costs, undesirable interactions between trees and livestock etc. so that the net reward of switching from conventional farming to agroforestry is not apparent. Consequently, without successful local examples or reliable assessments of expected benefits, farmers are often hesitant to take the risk of adopting agroforestry practices.

Our aim is to narrow down the uncertainties linked to adopting agroforestry and thereby support farmers in their management decisions. The approach we take is based on the methods of decision analysis. Taking a systems perspective, decision analysis goes beyond pure cost-benefit analysis by seeking to consider all system aspects that may be relevant to the farmer. Therefore, a first step is to collect knowledge on all potentially relevant factors based on literature reviews, expert interviews and/or own experiments. In a next step, farmers, other stakeholders and scientists collaborate to develop a conceptual model of the systems of interest. When translating this model into a simulation tool, the model parameters are supplied as probability distributions reflecting both epistemic uncertainty ('lack of knowledge') and aleatory uncertainty ('variability of the variable'). Using Monte Carlo simulations, this uncertainty is propagated to the model projections and returned as outcome distributions. This tool allows farmers to explore likely effects of alternative decisions on the criteria relevant to them.

We illustrate this approach using an ongoing project, where we work together with German farmers to build a decision tool on the question of whether to integrate poultry into existing apple fruit orchards. The basic conceptual model (Figure 1) of the expected net present value (NPV) consists of two main threads: the first covers the costs and benefits of the already established apple orchard, the second one the costs and benefits linked to the inclusion of chickens. Introducing chickens into the system, we expect a reduction in the costs to control voles, insect and apple scab. These can be seen as synergistic effects of the combined apple and chicken system. In a next step, this basic model will be extended by further factors identified to affect farmers' decisions such as labour requirements, legal hurdles and effects on environmental protection.

In the current state of the model, the results of the Monte Carlo simulations showed that in roughly 60% of the cases the NPV is positive when including chicken into the orchard. However, some uncertainty remains, which is indicated by a wide distribution of projected NPV values, which includes a considerable share of negative outcomes. A sensitivity analysis identified 'egg price' as by far the most influential variable affecting the NPV, suggesting that further research should focus on market trends to allow to further narrow down uncertainty and develop a robust decision tool.